

ARTICLE

Order matters: Autocorrelation of temperature dictates extinction risk in populations with nonlinear thermal performance

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Abstract

Forecasting the risks caused by climate change often relies upon combining species' thermal performance curves with expected statistical distributions of experienced temperatures, without considering the order in which those temperatures occur. Such averaging approaches may obscure the disproportionate impacts that extreme events like heatwaves have on fitness and survival. In this study, we instead incorporate thermal performance curves with population dynamical modeling to elucidate the relationship between the sequence of temperature events—driven by temporal autocorrelation—and extinction risks. We show that the permutation of temperatures determines the extent of risk; as thermal regimes grow warmer, more variable, and more autocorrelated, the risk of extinction grows non-linearly and is driven by interactions between the thermal distribution and its temporal autocorrelation. Given that the mean, variance, and autocorrelation of temperatures are changing in nuanced ways across the globe, understanding these interactions is paramount for forecasting risk. Using empirical data from a benchmarked set of thermal performance curves, we demonstrate how extinction risk is impacted by interacting changes to temperature's distribution and autocorrelation level, while controlling for seasonal and diurnal cycling. Our results and modeling approach offer new tools for testing the robustness of thermal performance curves and emphasize the importance of looking beyond temporally blind metrics, like mean population size or average thermal distributions, for forecasting impending extinction risks.

KEYWORDS

autocorrelation, colored noise, extinction risk, logistic model, population dynamics, stochasticity, thermal performance

INTRODUCTION

A species' sensitivity to changing thermal environments depends both on how temperatures are changing and on

its capacity to respond to those changes (Buckley et al., 2022; Deutsch et al., 2008; Huey et al., 2012). Different organisms, populations, and species have vastly different thermal sensitivities, dependent on the environments

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they have experienced, developed in, and evolved under (Kingsolver et al., 2013; Munday et al., 2013; Sinclair et al., 2016). The life or death of an individual—and the persistence or extinction of a broader population—is thus intrinsically dependent on its thermal history. Even brief extreme temperature events can have disproportionately negative impacts on survival; whether such events occur far apart, such that organisms can heal and populations can recover, or close together, such that the stress of one event compounds that of another, leads to vastly different outcomes. Consequently, estimating thermal risk without accounting for the explicit ordering in which temperatures occur may not accurately capture the effects of increasing thermal variability and warming.

An increasingly popular method of forecasting thermal risk in ectothermic species predicts long-term average performance or fitness by integrating a thermal performance curve (TPC) over a projected frequency distribution of temperatures (e.g., Figure 1a; Deutsch et al., 2008; Duffy et al., 2022; Huey & Stevenson, 1979; Pinsky et al., 2019; Tewksbury et al., 2008; Vasseur et al., 2014). TPCs are representations of the physiological function of temperature-dependent traits (e.g., maximum running speed, reproductive output; Angilletta, 2009); they have been measured for a variety of performance traits across thousands of species (Dell et al., 2013; Kontopoulou et al., 2024) and can be used to predict population-level rates (like growth or decline) at those same temperatures for ectothermic organisms (Amarasekare & Savage, 2012; Vasseur, 2020)

(though see Levesque and Marshall (2021) for their applicability to endotherms).

Although useful, this approach of forecasting risk by simply comparing TPCs with distributions of projected temperatures is based on a mathematical averaging that does not account for the outsized impact catastrophic events like heatwaves have on individuals or populations. For example, a temporal sequence of highly stressful temperatures, like a series of exceptionally hot days during a heatwave, could cause death or extinction despite generating only a small reduction in the calculated long-term average fitness (e.g., red temperature time series and population trajectory compared to blue or gray in Figure 1b–e). While many improvements have been made to this method, such as incorporating high-resolution thermal data and adjusting TPCs to more accurately depict the severe consequences of extreme temperatures (e.g., Buckley et al., 2022; Kingsolver et al., 2013), its temporally blind nature ultimately limits its utility. Improving our TPC-based risk forecasting requires us to move beyond such assumptions.

At the population level, risk is often attributed to population dynamic phenomena such as decline or extinction (Lande, 1993; Thomas et al., 2004; Urban, 2015). Because population dynamics are inherently sensitive to the temporal sequencing or “autocorrelation” of environmental changes, merging TPCs into population dynamic models may provide a stronger basis for measuring risk than classical averaging approaches. Ecological theory has long recognized the

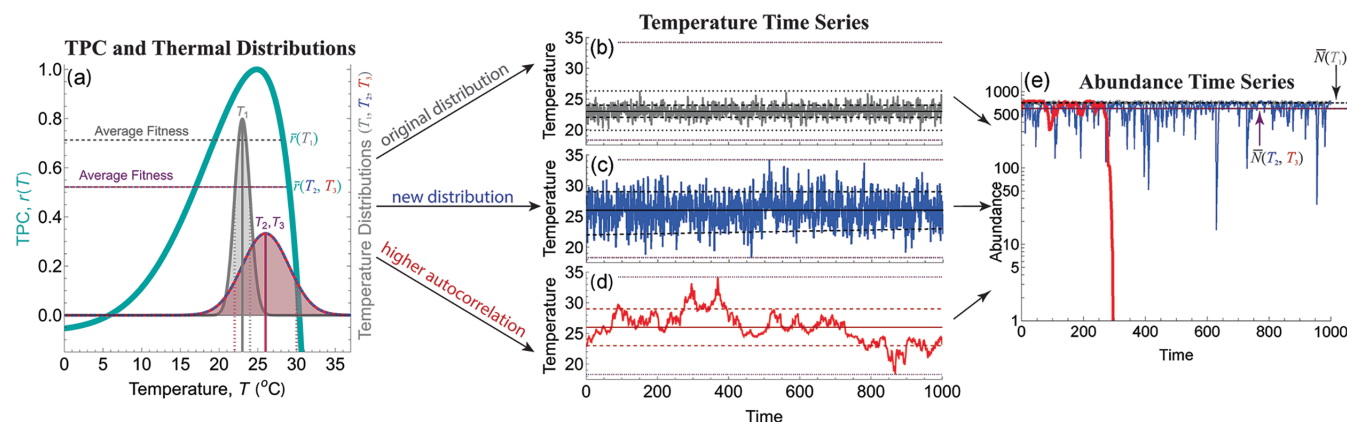


FIGURE 1 Comparison of risk forecasting methods using thermal performance curves (TPCs). (a) A normalized TPC of population growth rate $r(T)$ (teal) compared to several frequency distributions of temperatures (T_1 in gray, T_2 in blue, T_3 in red; latter two distributions are overlapping) and the average fitness predicted by integrating the TPC across them (dashed lines; $\bar{r}(T)$). (b) The first temperature time series T_1 (gray) is normally distributed according to $\mathcal{N}(23, 1)$ and not autocorrelated through time (such that the spectral exponent $\gamma = 0$; see Methods), while (c) T_2 (blue) has a shifted distribution $\mathcal{N}(26, 2.5)$ under no autocorrelation ($\gamma = 0$) and (d) T_3 (red) has an identical distribution $\mathcal{N}(26, 2.5)$ but a high level of autocorrelation ($\gamma = 2$). (e) The resulting abundances (solid lines) compared to the predicted abundances (straight dashed lines; $\bar{N}(T)$) show that while the decline in average fitness predicts the reduced abundance imposed by the shifted distribution of T_2 , it cannot predict the extinction risk posed by the high autocorrelation of T_3 .

potential for environmental temporal autocorrelation to increase extinction risk by clumping harmful events together (Lawton, 1988; Schwager et al., 2006; Vasseur & Yodzis, 2004). Physiologists have similarly found that temporal autocorrelation increases the sensitivity of ectothermic organisms to those stressful conditions (Jørgensen et al., 2019; Rezende et al., 2014, 2020; Sinclair et al., 2016). In much of the world, projected intensifications of heatwave regimes exacerbate such circumstances; using time-frequency decomposition of CMIP6 models, Duffy et al. (2022) found increasing temporal autocorrelation at 60% of aquatic and 80% of terrestrial locations (excluding Antarctica). The rising global incidence of stressful temperatures, where fitness and performance precipitously decline (Jørgensen et al., 2022), heightens the potential for warming thermal regimes to increase extinction risk in a manner dependent on the order in which those temperatures occur. Because current risk estimates broadly exclude explicit thermal regimes and their corresponding population dynamics, they are likely underestimating extinction and extirpation risks of species experiencing stressful, variable, and highly autocorrelated thermal regimes.

In this paper, we test the importance of incorporating population dynamics into forecasting models. While Duffy et al. (2022) offered an initial assessment of how population dynamics compare between historic and forecasted thermal regimes, they do not explicitly examine the effects of varying temporal autocorrelation. We advance this work by testing how population size, extinction risk, and subsequent risk metrics respond to variation in temporal autocorrelation across a range of otherwise statistically identical thermal regimes. We then examine the repercussions of varying the autocorrelation levels of the observed thermal regimes experienced by 38 invertebrate species to demonstrate how autocorrelation has modulated risk over time.

METHODS

To explore the relationship between temporal autocorrelation and thermal risk, we used a temperature-dependent logistic growth model to track population abundance and extinction events throughout carefully varied temperature time series. We first tested how changing the mean, standard deviation (SD), and temporal autocorrelation of normally distributed thermal regimes affected the population dynamics of example thermal generalists and specialists (Model 1). We then tested this modeling approach on more realistic thermal distributions and tolerances by using empirical TPCs from a 38-species insect dataset, varying the autocorrelation levels of observed temperature time series

from their collection locations (Model 2), and comparing simulation outcomes to various extinction risk metrics.

Population dynamics

We incorporated thermal sensitivity into population dynamical models using the intrinsic population growth rate (or Malthusian fitness) r . Defined as the difference between per capita birth and death rates ($r = b - d$), population growth rate has a well-established dependence on temperature (Amarasekare & Savage, 2012; Fey et al., 2021; Vasseur, 2020) which follows a unimodal and left-skewed TPC for most ectotherms (Angilletta, 2009; Dell et al., 2011; Rezende & Bozinovic, 2019). Many studies use $r(T)$ as a proxy to describe the relationship between individual performance or population fitness and temperature (e.g., Deutsch et al., 2008; Kingsolver, 2009; Thomas et al., 2012). Here, we incorporated a fitness TPC ($r(T)$) into the logistic growth model to describe the instantaneous rate of population growth or decline at any given temperature T .

This temperature dependence of population fitness has been described by a wide variety of functions with a myriad of complex parameterizations (see e.g., Kontopoulos et al., 2024; Padfield et al., 2021). Any formulation of a fitness TPC $r(T)$ with a similar shape will generate the same dynamics, so for simplicity (and following similar methods to Amarasekare & Savage, 2012; Thomas et al., 2017), we used a generic version arising from the difference between a unimodal birth function (as in Amarasekare, 2015, 2024, and references therein) and an exponentially increasing death function (as in Gillooly et al., 2001, 2002; Brown et al., 2004; Savage et al., 2004):

$$r(T) = b(T) - d(T) = \left(e^{-\left(T - b_{\text{opt}} \right)^2 / b_b} \right) - \left(m_a e^{(m_b T)} + m_c \right), \quad (1)$$

where the birth function is parameterized by b_{opt} (the optimal reproductive temperature) and b_b (its thermal breadth) and the death function is parameterized by m_a , m_b , and m_c (a scaling factor, a rate factor, and an additive constant, respectively).

We then calculated population sizes using the closed-form solution of the r - α logistic growth equation:

$$\frac{dN}{dt} = N(r(T) - \alpha N) \quad (2)$$

where N is the population size and T the temperature at time t . We used this formulation of the logistic growth model instead of the typical r - K formulation due to the

relative ease with which it handles negative fitness values (which importantly indicate population decline) and because it assumes a constant strength of density dependence α rather than a constant carrying capacity K (Long et al., 2007; Mallet, 2012; Vasseur, 2020); one can easily convert between the two by substituting $K = r/\alpha$. Because abundance in the logistic model declines asymptotically to 0 when $r < 0$, we defined an extinction threshold for our simulations as $N < K \times 10^{-6} = 1/\alpha \times 10^{-6}$. Shown results use $\alpha = 0.001$, but changing the value of α impacts only the magnitude, not the behavior, of model dynamics (see Appendix S1).

For Model 1, we used Equation (1) to parameterize two generic TPCs: one for a “generalist” species, with a cooler thermal tolerance and a wider thermal breadth (Figure 2a, teal; with peak performance at the thermal optimum $T_{\text{opt}} = 25^\circ\text{C}$, thermal limits at the thermal minimum $T_{\text{min}} = 5.8^\circ$ and maximum $T_{\text{max}} = 30.4^\circ$, and positive growth rates over a thermal breadth $T_b = 24.6^\circ$), and one for a “specialist” species, with a warmer thermal tolerance but a narrower thermal breath (Figure 2a,

purple; $T_{\text{opt}} = 28^\circ$, $T_{\text{min}} = 14.5^\circ$, $T_{\text{max}} = 32^\circ$, and $T_b = 17.5^\circ$). To emphasize the generality of our approach, neither curve is linked to a specific study organism, but both are similar to several species from the oft-used Frazier et al. (2006) dataset (see Appendix S2), which we examine in detail in Model 2.

Autocorrelated temperature variation

To determine the importance of incorporating temporally explicit modeling into risk assessments, we compared populations experiencing the same thermal regimes but different amounts of autocorrelation. Temporally blind metrics, or ones that do not take the ordering of temperatures into account, always predict the same level of risk for thermal regimes with the same statistical moments. Autocorrelation, however, may meaningfully impact the realized risk because increasing autocorrelation “clumps” similar temperatures together within a series (Lawton, 1988; Schwager et al., 2006; Vasseur & Yodzis, 2004), leading to

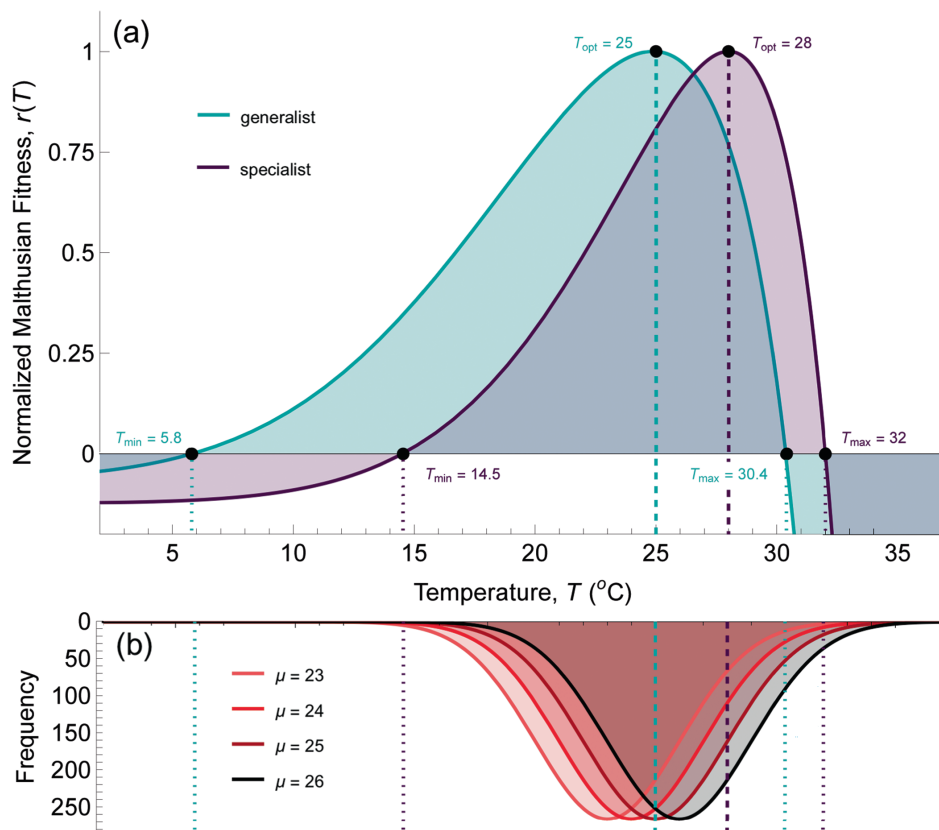


FIGURE 2 (a) Normalized sample thermal performance curves (TPCs) for two species, a generalist (teal; Equation 1 parameters $b_{\text{opt}} = 29$, $b_b = 180$, $m_a = 5 \times 10^{-7}$, $m_b = 0.4$, and $m_c = 0.05$, normalized by a factor of 0.755) and a specialist (purple; $b_{\text{opt}} = 35$, $b_b = 170$, $m_a = 5 \times 10^{-7}$, $m_b = 0.4$, and $m_c = 0.05$, shifted 2 degrees to the right relative to the generalist, normalized by a factor of 0.4067), each annotated with their thermal optima (T_{opt} , dashed lines), minima (T_{min}), and maxima (T_{max} ; dotted lines). (b) The thermal distributions used in Model 1 (shown with SD $\sigma = 3^\circ$ and mean temperatures indicated by μ).

prolonged heatwaves or cold spells that could cause large differences between temporally blind and temporally explicit risk assessments.

To generate autocorrelated time series of temperature variation, we used a technique known as **spectral synthesis** (Cuddington & Yodzis, 1999; Fontaine & Gonzalez, 2005; Halley et al., 2004). This technique generates time series whose power spectra are characterized by a $1/f^\gamma$ scaling property, where f is the frequency of the noise and $0 \leq \gamma \leq 2$ is the spectral exponent (Halley, 1996; Vasseur & Yodzis, 2004). The spectral exponent characterizes the “color” of noise, where white noise ($\gamma = 0$) represents no autocorrelation and includes an equal mixture of all frequencies (such that fluctuations lack frequency dependence altogether), brown noise ($\gamma = 2$) represents high autocorrelation and has variance driven mostly by low frequencies (also known as a random walk or Brownian motion), and pink noise ($\gamma = 1$) lies in between (Halley, 1996; Vasseur, 2007). For a given spectral exponent γ , we generated a zero-mean normally distributed time series Y_γ from $t = 1, \dots, n$ by summing $f = 1, \dots, n/2$ sine waves:

$$Y_\gamma(t) = \sum_{f=1}^{n/2} f^{-\gamma/2} \sin[2\pi ft/n - \theta_f], \quad (3)$$

where θ_f is a random vector of phase operators drawn from the uniform distribution $[0, 2\pi)$ and the amplitude $f^{-\gamma/2}$ generates the power-law scaling (Vasseur, 2007). In practice, computing the sums in Equation (3) is inefficient and the same result can be obtained using an inverse fast Fourier-Transform (Cochran et al., 1967; implemented herein using the “InverseFourier” function in *Mathematica*, also available through the inverse “fft” function in *R*).

To provide greater insights into the role of temporal autocorrelation on model dynamics, we used an algorithm known as **spectral mimicry** (Cohen et al., 1999) to apply the synthesized random temporally autocorrelated sequence to a conserved set of temperature values. For each combination of temperatures with mean μ and SD σ , we generated a normally distributed reference series $X_{\mu,\sigma}$ of n temperatures by back-sampling values from the cumulative distribution function at $n + 1$ regularly spaced intervals. This creates a “perfect” distribution of temperature values that we permuted according to the ordinal ranking of values in Y_γ . This permutation method imparts the desired autocorrelation structure in Y_γ onto $X_{\mu,\sigma}$ while preserving the same mean and variance (and, indeed, values) across different levels of autocorrelation. While the spectral exponent of the permuted sequence can differ slightly from that of Y_γ , the difference is generally very small and offset by the benefit of using the same

values at each level of autocorrelation. Spectral mimicry thus ensures that comparisons between sets with the same temperature distribution but different spectral exponents (e.g., blue and red series in Figure 1c,d) reflect only differences in the ordering of the temperatures, not differences in the temperatures themselves.

Model 1: Simulating risk across thermal regimes

To assess how temporal autocorrelation modulates risk, we simulated population dynamics for a generalist and a specialist species (Figure 2a) across normally distributed temperatures with different combinations of means (ranging from $\mu = 23$ to 26°C at intervals of 0.5°), SDs (ranging from $\sigma = 1$ to 4° at intervals of 0.5°), and autocorrelation levels (ranging from $\gamma = 0$ to 2 at intervals of 0.1). We ran 1000 replicates of simulations lasting 2000 time steps for each thermal regime and TPC, with outputs for each parameter combination of (1) the number of populations to go extinct and (2) the time step of each extinction. Simulations are additionally capable of storing (3) the population size during each time step; however, as this is significantly more computationally intensive, population sizes were only retained for a select parameter space of interest (four evenly spaced mean temperatures at $\sigma = 3^\circ$ across all autocorrelation levels; Figure 2b). The finite number of temperatures included in each run required us to choose an appropriate time step length for analyzing the dynamics produced from our ordinary differential equation; relative to the generation time, overly long time steps cause population dynamics to average across all variation, while overly short ones result in population dynamics closely tracking all variation, regardless of the actual dynamics. As our TPCs were normalized with $r \leq 1$, we used a time step size of 1 where neither averaging nor tracking dominates (see Appendix S3). Each simulation began with an initial population size equal to the average population size expected for that thermal regime, such that $N_0 = \bar{r}(X_{\mu,\sigma})/\alpha$. All models were run in *Mathematica* V.13.0.1; the essential functions are also available in *R* V.4.2.0.

Model 2: Empirical TPCs and climate data

To assess autocorrelation’s effects on more realistic temperatures, we simulated population dynamics across different permutations of observed climate data from a global dataset of 38 insect species’ intrinsic growth rate TPCs (gathered from 31 studies and 35 locations between 1974 and 2003), assembled in Frazier et al. (2006) and

previously analyzed in Deutsch et al. (2008), Vasseur et al. (2014), and Duffy et al. (2022). For comparison, we analyzed these empirical data using the same Gaussian-quadratic TPC fit as in earlier papers (see Appendix S2), allowing negative growth rates at high temperatures following Vasseur et al. (2014) and Duffy et al. (2022). To test the impact of changing autocorrelation levels on realistic thermal regimes—which are rarely perfectly normally distributed—we gathered a decade of observed historical (January 1, 1994–December 31, 2003) and recent (January 1, 2014 to December 31, 2023) daily minimum and maximum temperatures from each species' collection location using Visual Crossing's Weather Data Services (7300 time steps per decade and location; see Appendix S4). As a benchmark, we first measured the observed spectral exponent of each unedited climate dataset. Then, instead of artificially varying the thermal regime's mean and SD (as in Model 1), we conserved the *values* of the meteorological datasets while permuting their *ordering*. We stochastically generated 1000 replicates of each distribution (historical or recent) at each autocorrelation level (spectral exponents ranging from $\gamma = 0$ to 2 at intervals of 0.1), then simulated the resulting deterministic population dynamics for each of those permutations. Because TPCs were normalized and this dataset included 2 data points per day, for this model we used a time step size of 0.5. Simulations were run in *Mathematica* V.13.0.1.

Unlike in the synthetic temperature time series used in Model 1, scrambling meteorological data to vary the level of autocorrelation requires additional methods to conserve seasonal and diurnal fluctuations. Retaining those fluctuations in permuted observational datasets is critical to ensuring that our spectral methods did not disrupt realistic (and likely relevant) elements of the climate data, as well as to test whether changes to the temporal autocorrelation structure are buffered by seasonal and diurnal cycling. To maintain seasonality but alter autocorrelation, we first seasonally detrended the climate data by identifying and subtracting out the seasonal fluctuation from each dataset. We then used spectral synthesis and mimicry to manipulate its spectral exponent (as above) before adding the identified seasonal trend back in. This procedure allowed us to maintain the “power” ratio of seasonal to non-seasonal components, the former of which was preserved completely and the latter of which varied between autocorrelation treatments. Within this process, we retained diurnal fluctuations by applying spectral mimicry to the *average* temperature of each day (instead of the collected minima and maxima), such that the days' average temperatures were sorted with the designated autocorrelation level, but the actual temperature time series included the corresponding extremes.

Notably, we assumed a static TPC for each species reasonably approximated their population growth rate at any point during a year; this is not realistic, as insects generally have multiple life stages timed with seasonality (particularly in temperate climates). While a detailed, species-specific model should unequivocally take more careful steps to incorporate realistic life history traits (particularly the temporal components and differential thermal sensitivities of fecundity, mortality, and development; see Amarasekare and Savage (2012) and Kingsolver and Buckley (2020) for more nuanced discussions), the approach used herein is simply intended to examine whether realistic TPCs paired with realistic climate data yield risk forecasts responsive to changes in temporal autocorrelation. Although these results should thus not be interpreted as literal extinction risk forecasts for each species, this approach does represent significant progress beyond classical methods (which also assume constant population growth according to TPCs, but often do not incorporate seasonality into the climate data) and follows similar approaches from recent literature (e.g., see methods in Duffy et al. (2022): Population Dynamical Modeling).

Risk metrics

To assess the effects of autocorrelation on the accuracy of risk forecasting, we compared temporally explicit simulation outcomes with several temporally blind metrics. The temporally blind metrics included the warming tolerance (WT; the difference between $T_{\max}$ and the mean environmental temperature μ) and the thermal safety margin (TSM; the difference between T_{opt} and the mean environmental temperature μ) (Deutsch et al., 2008). Many other risk metrics rely instead on projected population size; for example, the IUCN Red List categorizes species as Vulnerable, Endangered, or Critically Endangered if they meet the risk criteria for any one of five categories (A. Population size reduction, B. Geographic range reduction, C. Small population size and decline, D. Very small or restricted population, or E. Quantitative Analysis), four of which explicitly base categorizations on population size and one of which (B) is often a risk factor *because* of its impact on population sizes (IUCN, 2012). To calculate the projected population size in a temporally blind fashion, we took the average fitness expected for a thermal regime $X_{\mu,\sigma}$ divided by the density dependence parameter α .

The temporally explicit outcomes were average abundance, population stability (both calculated across the time series of population size up until extinction occurs), the timing of extinction events, and the proportion of

extinct runs. We calculated population stability using the consecutive disparity index (D), which quantifies the mean rate of change between each pair of neighboring values p_i and p_{i+1} across a series of length n :

$$D = \frac{1}{n-1} \sum_{i=1}^{n-1} \left| \ln \frac{p_{i+1}}{p_i} \right| \quad (4)$$

and is thus better able capture the effects of changing autocorrelation than other common metrics like the coefficient of variability (Fernández-Martínez et al., 2018). Higher values of D indicate greater changes between consecutive values of the series and thus lower stability, whereas lower values of D correspond to higher stability.

In Model 1, the smaller dataset (two TPCs) allowed us to explore the dynamics of all temporally blind and temporally explicit metrics. In Model 2, the larger

dataset (38 TPCs) required a more constrained analysis focused only on whether risk increased across the compared decades. We thus included the temporally explicit metrics of time to extinction and proportion of extinct runs, as well as (in the case of runs that did not experience any extinction risk) the minimum observed population size.

RESULTS

Effects of autocorrelation across the thermal landscape

Model 1 demonstrated that the effects of changing thermal distributions (mean, SD) on extinction risk are mediated by the level of temporal autocorrelation. Figure 3a shows the proportion of extinct runs across the parameter

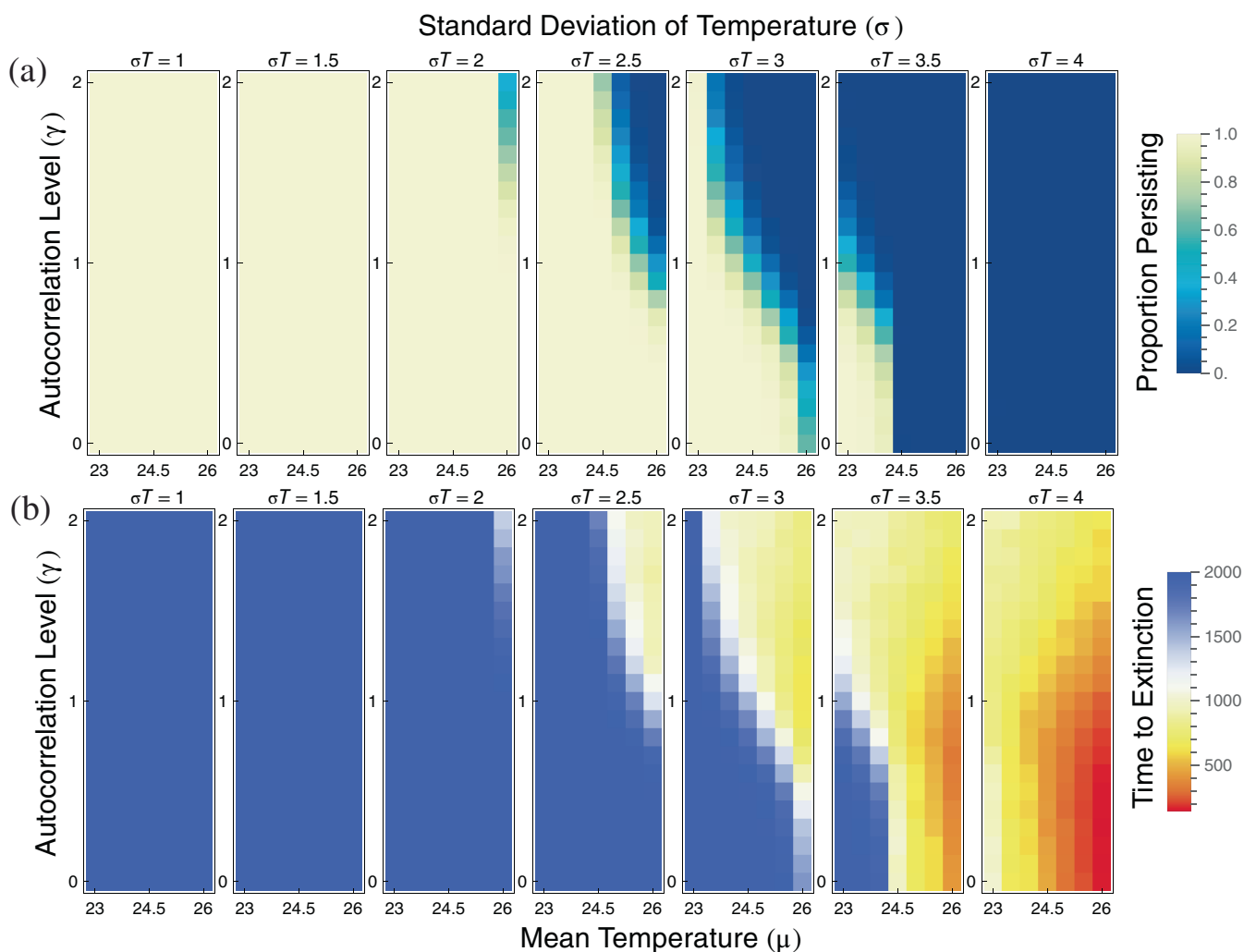


FIGURE 3 The extinction outputs of Model 1 for the generalist thermal performance curve (TPC), showing (a) the proportion of extinct runs and (b) the average extinction time (where an extinction time of 2000 (blue regions) is equivalent to no extinction events occurring) across autocorrelation levels (γ) and mean temperatures (μ) at each SD (σ) over 1000 replicates lasting 2000 time steps.

space for the generalist TPC (with tan indicating persistence and dark blue indicating extinction), with each subpanel showing the relationship between mean temperature and temporal autocorrelation at a different SD (expanded plots and results for the specialist species, which are qualitatively identical, are shown in Appendix S5). As expected, permissive thermal regimes (mean temperatures within the thermal breadth, low variability, and low autocorrelation) resulted in no extinction risk, and increasing any individual parameter increased risk (trends from tan to blue moving up and right within and across Figure 3a subpanels). However, the relevant magnitude of change varied between parameters. Changing the level of autocorrelation had no effect when the SD was too low ($\sigma < 2^\circ$) or too high ($\sigma > 3.5^\circ$); in those cases, the overall variation is either so low that extinctions cannot be generated regardless of the temperatures' ordering or so high that even randomly scattering the extremes throughout the simulations cannot save the population. Increasing autocorrelation had the most pronounced impacts on extinction risk in mid-dling thermal regimes, where $2^\circ \leq \sigma \leq 3.5^\circ$ (for hotter mean temperatures ($\mu \geq 25^\circ$) as $\sigma \rightarrow 2^\circ$ and cooler mean temperatures ($\mu \leq 24^\circ$) as $\sigma \rightarrow 3.5^\circ$).

Autocorrelation also impacted the timing of extinctions. Figure 3b shows the same set of simulations as Figure 3a, but now colored by the average extinction time. Interestingly, while higher autocorrelation resulted in a higher *proportion* of extinctions, it did not uniformly result in *earlier* extinctions. Instead, lower levels of autocorrelation corresponded to faster extinctions under the most stressful thermal regimes (red coloration concentrating in the bottom right of Figure 3b subplots for $\sigma \leq 3.5^\circ$). While the same total number of extinctions occurred across autocorrelation levels for these highly stressful thermal regimes, the combination of hot mean temperatures and large SDs affected the type of event causing those extinction. When individual extreme temperature events are hot enough, the corresponding population decline becomes so precipitous that scattering each individual extreme temperature randomly throughout the simulation (white noise, $\gamma \rightarrow 0$) is riskier than clumping them all together (brown noise, $\gamma \rightarrow 2$) because any extreme temperature on its own could cause extinction. In other words, when individual heat events become extremely deadly, high autocorrelation lengthens average persistence times simply by randomly clumping the extremes together later in some simulation runs.

Comparing risk metrics

To establish the more detailed impacts of autocorrelation on specific risk metrics, we retained the population sizes

(as well as the extinction outcomes) across simulations for the parameter space where autocorrelation had the most distinctive effect on extinction outcomes ($\mu = 23, 24, 25$, and 26°C , $\sigma = 3^\circ$) and compared the outcomes to the temporally blind metrics. At these mean temperatures, the generalist ($T_{\text{opt}} = 25^\circ$, $T_{\text{max}} = 30.4^\circ$) had warming tolerances (WTs) of 7.4, 6.4, 5.4, and 4.4° and thermal safety margins (TSMs) of 2, 1, 0, and -1° , respectively, at every level of variance and autocorrelation. The specialist ($T_{\text{opt}} = 28.3^\circ$, $T_{\text{max}} = 32^\circ$) had WTs of 9, 8, 7, and 6° and TSMs of 5.3, 4.3, 3.3, and 2.3° , respectively, at every level of variance and autocorrelation. These WTs and TSMs indicate a trend of declining resilience with increasing temperatures, but suggest potential risk only for the generalist TPC and only at the hottest two temperatures.

Figure 4 shows the simulated average population abundance, stability, and persistence probability for these same parameters across autocorrelation levels, with results for the generalist TPC on the left and the specialist TPC on the right. Temporally blind projections of population sizes (dotted lines, Figure 4a,d) were all positive, with the highest population sizes projected successively from coolest to warmest means for the generalist TPC and from warmest to coolest means for the specialist TPC. The temporally explicit modeling followed the same trends. However, while the temporally blind projection closely predicted the average abundance of the generalist species, it overpredicted specialist abundance at all temperatures. This is a consequence of Jensen's inequality, whereby the asymmetry of the TPC (which is steeper and narrower for the specialist) results in thermal variability disproportionately decreasing abundances at high temperatures beyond what is predicted by the mean conditions (Bernhardt et al., 2018; Martin & Huey, 2008). For both TPCs, varying the autocorrelation levels had very little effect on abundance beyond increasing its variability between runs (more so for the generalist TPC and hotter mean temperatures; growing uncertainty moving right in Figure 4a,d).

Stability declined universally with increasing mean temperatures and with decreasing levels of autocorrelation (Figure 4b,e). The shown metric of stability (the consecutive disparity index D ; other metrics are shown in Appendix S5: Figure S5) indicates the average rate of change between consecutive population sizes. Its decline with hotter mean temperatures (higher values of the darker lines in Figure 4b,e) results from the steeper slopes of the TPCs at those temperatures: in these regions, each unit change in temperature has a greater effect on $r(T)$, which precipitates more rapid changes in population size between time steps (more so for the specialist than for the generalist, again due to the comparatively steeper decline of its TPC; see Figure 2a). The

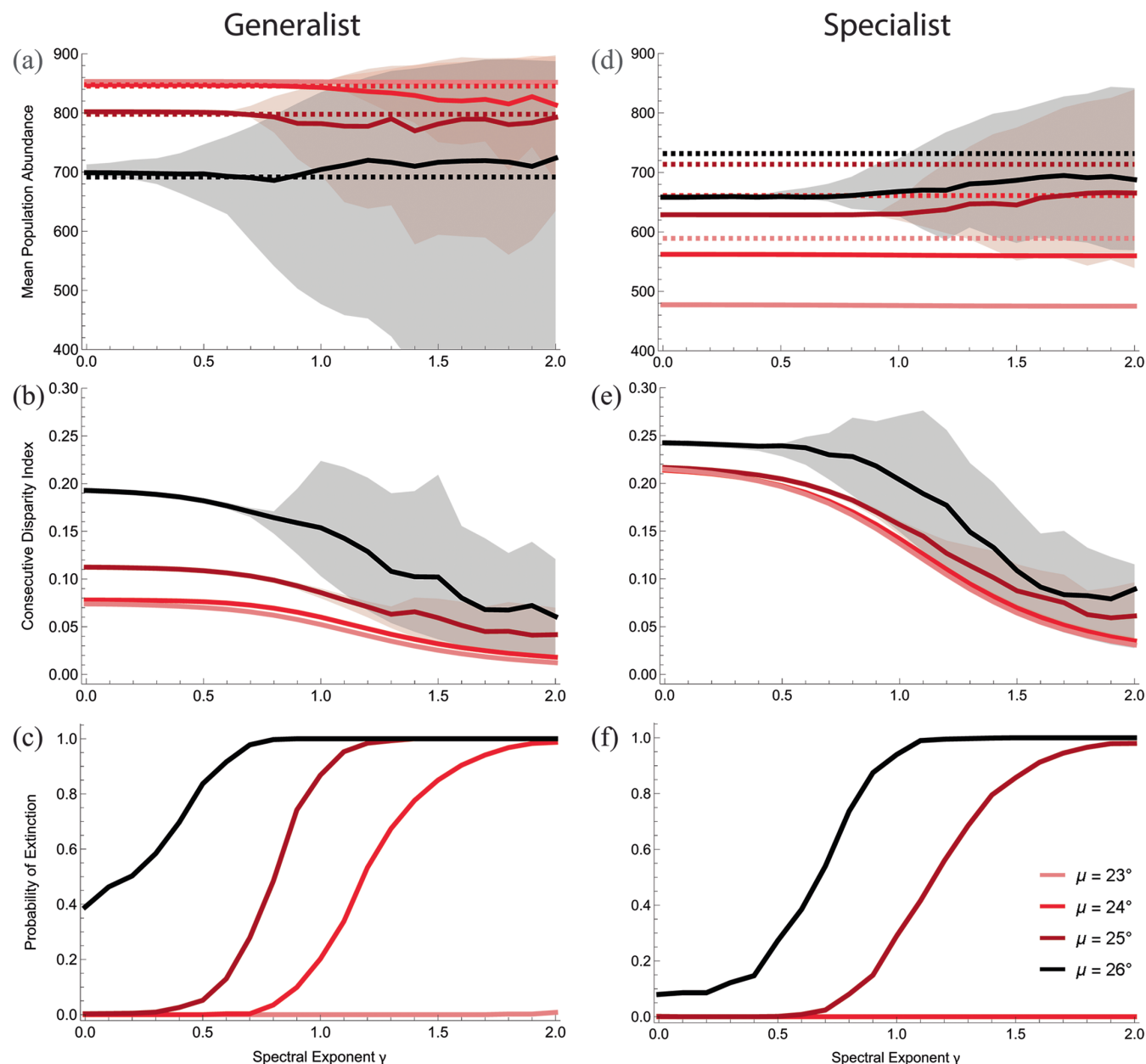


FIGURE 4 The population size outputs of Model 1, with results for the generalist thermal performance curve (TPC) on the left and the specialist TPC on the right. Darker colors indicate higher mean temperatures (μ). Shaded regions indicate the 10th–90th quantile of the given metric. (a, d) Mean abundance plots show the simulated mean abundance up to the time of extinction (solid lines) and the predicted mean abundance for that thermal regime (dotted lines). (b, e) Stability plots show the consecutive disparity index D up to the time of extinction, with lower values indicating higher stability. (c, f) Extinction plots show the number of simulations out of 1000 where the population abundance dropped below the extinction threshold.

increase in stability at higher levels of temporal autocorrelation, on the other hand, follows from the sequencing of temperatures. The temperature change between consecutive time steps under white noise ($\gamma = 0$, left side of subplots in Figure 4) is random and may thus be quite large, whereas under brown noise ($\gamma = 2$, right side of subplots in Figure 4), similar values are expected to be clumped together, leading to relatively smaller changes in the environmental conditions (and thus the resulting

changes in population size) between consecutive time steps.

For both TPCs, the actual probability of extinction increased with both the mean temperature and the autocorrelation level (Figure 4c,f). The generalist population faced higher extinction risk than the specialist population across the board, though at the hottest two mean temperatures both faced nearly a 100% chance of extinction at high levels of autocorrelation. While autocorrelation had

no impact on extinction risk under completely permissive thermal regimes ($\mu = 23^\circ$ for the generalist, $\mu = 23, 24^\circ$ for the specialist), it had a pronounced impact on extinction risk under more stressful ones; for example, at $\mu = 25^\circ$, the generalist species shifts from an extinction risk near 0% to an extinction risk near 100% in the range of $0.5 < \gamma < 1$.

Compared to the simulated outcomes, population abundance (both the temporally blind prediction and the actual average across simulations) was a poor predictor of risk: all mean temperatures and autocorrelation levels resulted in simulated mean abundances far above the extinction threshold, even when all populations went extinct under those conditions. For the specialist population, the abundance trends were also the inverse of the extinction trends: those with the smallest mean abundances ($\mu = 23, 24^\circ$) corresponded with the lowest extinction risks. Population stability was a good predictor of the risk imposed by warming, but not of the risk imposed by autocorrelation; stability universally increased under cooler temperatures and higher autocorrelation, whereas extinction risk universally increased under hotter temperatures and higher autocorrelation.

Impacts of autocorrelation on global risk assessments

Model 2 demonstrated that extinction risk remains responsive to temporal autocorrelation under realistic conditions. Using real thermal distributions and including diurnal and seasonal variation (e.g., Figure 5a), we found the same S-shaped relationship between extinction risk and autocorrelation level observed using the sample thermal performance curves and normally distributed temperatures (Figure 5b, compared to Figure 4c,f). These fluctuations did buffer risk somewhat by forcing nightly and seasonal relief from stressful temperatures, but we could recover the same S-shaped curve for every species by varying the extinction thresholds (for example, given a TPC, increasing the extinction threshold from $K \times 10^{-6}$ individuals to $K \times 10^{-3}$ for more permissive thermal regimes or decreasing it to $K \times 10^{-9}$ for more stressful ones). The remainder of the analysis was performed with an extinction threshold of 1 individual.

Projected species-level impacts

The output for each of the 38 species incorporates three results: whether spectral exponents increased or decreased over time, whether background thermal regimes made extinction more or less likely across all

autocorrelation levels, and how the probability of extinction across simulations changed based on observed spectral exponents. Detailed results, including a summary plot (Appendix S1: Figure S1), the measured climatic variables (historic and recent μ , σ , and γ) at each location (Appendix S1: Table S1), and each species' TPC, histogram of historic versus recent temperatures, and extinction curves (Appendix S1: Figure S2) are found in Appendix S6.

The observed level of autocorrelation increased at 23 of the 34 collection locations (67.6%; red points to the right of pink points in Figure 5b and Appendix S6: Figure S2, right subplots). The recent temperature distributions were more stressful than the historic temperature distributions for 16 species (42.11%; black line to the left of gray line in Figure 5b and Appendix S6: Figure S2), less stressful for 4 species (10.53%; black line to the right of gray line), and equally stressful for the remaining 18 species (47.37%; black and gray lines overlapping). Regardless of whether the spectral exponent itself increased or decreased between time periods, extinction risk increased exclusively in scenarios that became more sensitive to autocorrelation (extinction curves shifting left; e.g., Figure 5b, Appendix S6: Figure S1).

Across the 1000 simulations performed at the level of autocorrelation observed in the original dataset (i.e., runs where the original temperature distribution was stochastically permuted into 1000 different realizations with the same level of autocorrelation), extinction risk increased for 13 species (34.21%) and decreased for 3 species (7.89%; Figure 5c, purple regions). Whether the simulated extinction risk increased is determined by the combination of changes to the thermal distribution (curve shift) and the autocorrelation level (point shift); the thermal distribution determines the possible potential outcomes for a species with a given TPC, while the autocorrelation level determines where on that curve the species currently sits (or historical sat). The simulated chance of extinction remained unchanged for the remaining 22 species (57.89%); of those 22, 11 went extinct in every simulation (Figure 5c, orange regions) while 11 never went extinct at all (Figure 5c, yellow regions; see below).

To assess whether risk increased for those species whose probability of extinction was 100% in both historic and recent simulations, we looked at the mean time to extinction across simulations at each autocorrelation level (Figure 5c, orange regions; full results in Appendix S6: Figures S1b and S3). As expected from earlier modeling results, high levels of autocorrelation incrementally increased average persistence times; by clumping stressful conditions later in runs. However, at the measured spectral exponents, populations went extinct faster under recent conditions in 8 out of 11 scenarios (average decrease of

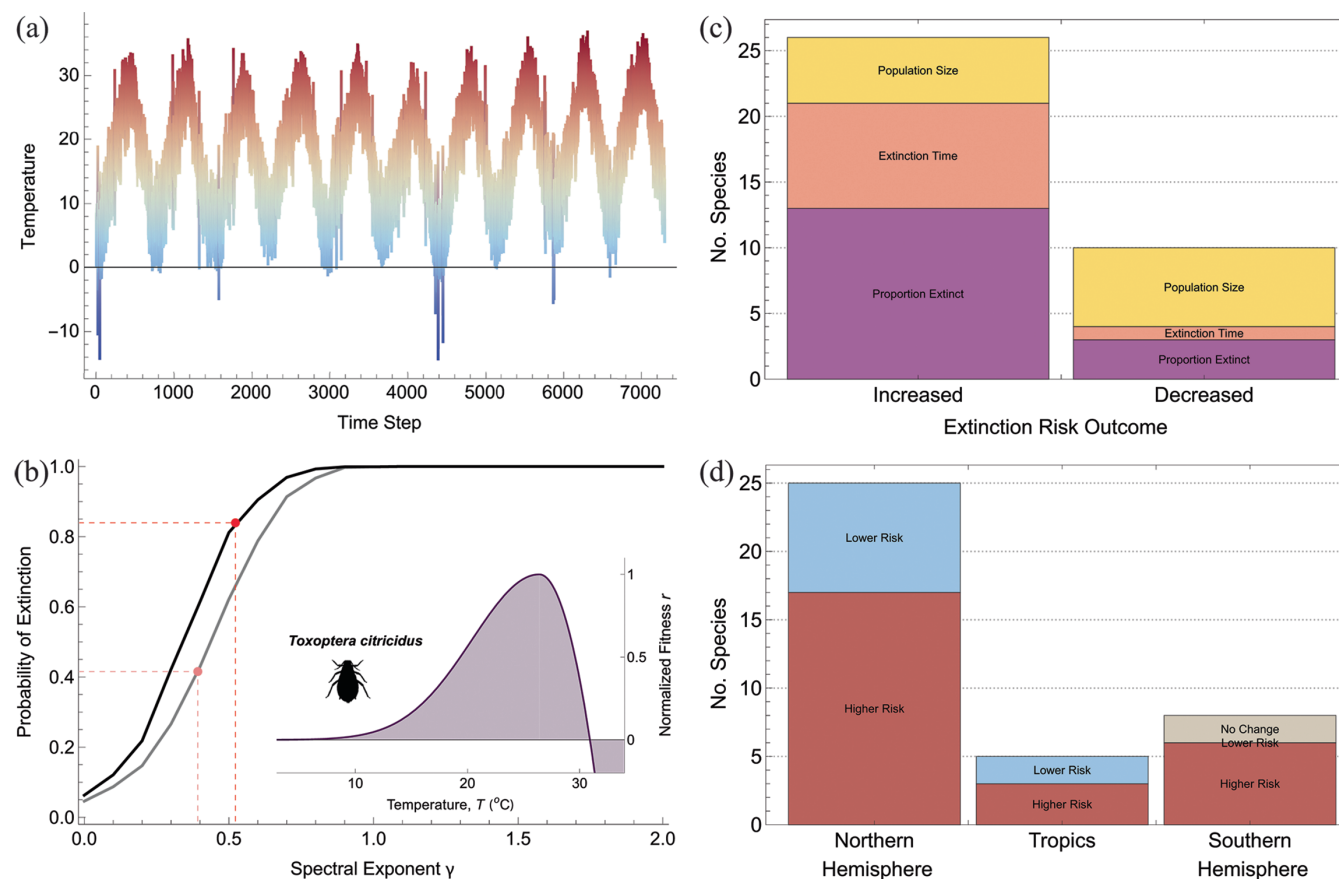


FIGURE 5 (a) Example temperature time series of the observed historical (1994–2003) temperatures at 34.98°N, 138.4°W, the collection location of *Toxoptera citricidus*, re-sorted with an autocorrelation level of $\gamma = 1$ with seasonal and diurnal fluctuations included. (b) Proportion of runs out of 1000 where *T. citricidus* went extinct at each autocorrelation level given observed historical (gray line, 1994–2003; $\mu = 16.48^\circ$, $\sigma = 9.08^\circ$) and recent (black line, 2014–2023; $\mu = 17.52^\circ$, $\sigma = 8.48^\circ$) temperatures and its empirical thermal performance curve (TPC) (purple subplot). Actual autocorrelation levels of the observed temperature time series, and their corresponding extinction risks, are marked in pink (historical; $\gamma = 0.39$) and red (recent; $\gamma = 0.52$). (c) The number of species whose projected extinction risk increased or decreased under recent (compared to historic) temperature conditions, divided up by the method used to assess risk: “Proportion Extinction” indicates a difference in the probability of extinction observed between simulation runs (as in (b)), while “Extinction Time” (used if all populations went extinct under both conditions) indicates whether that extinction occurred faster under recent versus historic conditions and “Population Size” (used if no populations went extinct under either condition) indicates whether the minimum population size was smaller under recent versus historic conditions. (d) The proportion of species sampled from each region whose projected risk increased or decreased. Insect silhouette “Aphididae” by Christoph Schomburg (July 14, 2020) courtesy of PhyloPic (license: CC0 1.0 Universal Public Domain Dedication).

57.71%, or 201 days); one species went extinct slower (increase of 2.99%, or 7 days); and two species always went extinct immediately, indicating that the regimes were too stressful for persistence under any circumstance.

For species whose probability of extinction was 0% under both historic and recent temperatures, we looked instead at the minimum population size encountered across simulations at each autocorrelation level (Figure 5c, yellow regions; full results in Appendix S6: Figures S1c and S4). We saw a distinct trend of declining minimum abundance with increasing spectral exponent in all cases. At the measured spectral exponents, minimum population sizes were smaller under recent

temperatures for 5 out of 11 species, but the difference between regimes where abundance increased was generally very small (average increase of 16 individuals) compared to those where abundance decreased (average decrease of 136 individuals).

In summary, 26 species faced higher risk under recent thermal regimes (68.42%) compared to 10 species with lower risk (26.32%) and 2 species with no change (5.26%). Fourteen species faced projected extinction risk over 90% under recent temperature scenarios. As shown in Figure 5d, increased versus decreased risk was divided 68%/32% in the Northern Hemisphere (25 species), 60%/40% in the tropics (5 species), and 75%/0% in the

Southern Hemisphere (8 species, with the remaining 25% facing 100% extinction risk across scenarios; shown in gray). The level of temporal autocorrelation affected extinction risk for 92% of tested species.

DISCUSSION

Our models show that the impact of changing thermal regimes on population persistence depends on the combination of parameters: increasing means and extremes do enhance extinction risks, but as conditions grow more stressful, that effect is increasingly mediated by temporal autocorrelation. Intensifying heatwave regimes vastly reduce predictability, stability, and persistence probability; those impacts grow more pronounced as the background thermal regime becomes more stressful. Indeed, while increasing a thermal regime's variance causes the pronounced increase in extinction risk expected from earlier results (e.g., Slein et al., 2023; Vasseur et al., 2014), this effect does not drive extinction risk alone unless the scenario is extremely permissive or extremely stressful. We further found that when experiencing a thermal regime stressful enough to generate extinctions but not stressful enough to guarantee them, realistic autocorrelation values often fell along the steep slope between probable persistence and probable extinction. While increasing means and variances of temperatures shape the background thermal landscape, the extent of autocorrelation fundamentally determines whether realistic thermal regimes are survivable or deadly.

Autocorrelation has previously been shown to increase extinction risk of populations in models where the environment additively changes the population growth rate (Petchey et al., 1997) or where it alters the carrying capacity of a population (Cuddington & Yodzis, 1999). Schwager et al. (2006) clarified that whether red noise increases or decreases extinction risk over white noise depends on the number of catastrophic events: redder noise decreases extinction risk when its “redness” ensures high similarity between values and thus reduces the likelihood of catastrophes. This distinction, while true, conflates the impact of higher autocorrelation with the impact of lower variance; our use of spectral mimicry has allowed us to show that when variance is held constant, red noise unambiguously increases extinction risk over white noise.

Here, we have incorporated temperature variation through a well-established and mechanistic non-linear relationship using TPCs, the shape of which incorporates the stressful impacts of heat events via the accelerating decline of the TPC at high temperatures (Figure 2). Autocorrelation is known to increase “tracking”—the relationship between the actual population size and the

equilibrium at any given temperature (Vinton & Vasseur, 2020)—because, for a given variance, changes at short timescales are smaller, allowing the population to more closely follow its equilibrium. When temperatures enter the stressful range in this model, $r(T) < 0$ and the equilibrium population size is 0; the relatively smaller variations at short timescales act to increase the length of stressful periods, increasing the likelihood that a population will reach critically low sizes and cross the extinction threshold. The ability of a population to “track” the equilibrium in our model depends on the magnitude of $r(T)$; as heat stress leads to more negative values of $r(T)$, the likelihood of extinction thus increases more quickly.

Standard risk metrics, like TSMs and population abundances, become significantly worse predictors of persistence under these high autocorrelation scenarios. Comparing predicted outcomes between species with a generalist TPC (more typical of a temperate climate) and a specialist TPC (more typical of a tropical climate) showed that hotter thermal regimes can increase specialist abundance and extinction risk simultaneously. However, the differential sensitivities of species to changes in temporal autocorrelation will likely depend on more than just the shape of their TPC; there are also correlations, for example, between faster life histories and greater sensitivities to the frequency of environmental variation (Paniw et al., 2018; Postuma et al., 2020; Smallegange et al., 2020; Tuljapurkar et al., 2009). It will be critical for scientists and policymakers alike to account for this aspect of climate change under warming scenarios with increasing autocorrelation, and may be particularly important in systems already known to experience higher levels, (e.g., marine environments; Duffy et al., 2022; Vasseur & Yodzis, 2004).

While this model provides the most comprehensive analysis of the interactions between changing thermal regimes, population dynamics, and population persistence to date, it includes a considerable number of assumptions. By choosing logistic growth of a density-dependent population, we have incorporated the built-in assumptions of that model, including that the thermal response of carrying capacity K will be the same as that for fitness r (Bieg & Vasseur, 2024). We have ignored ecological interactions (Dell et al., 2014; Fey & Vasseur, 2016) and assumed that empirical TPCs are a good proxy for population growth rates (Sinclair et al., 2016). Perhaps most critically, we have assumed that those TPCs are static across space and time: each individual in these models has the same thermal tolerance regardless of life stage (Kingsolver et al., 2011), thermal history (Rebolledo et al., 2021), population structure (Angert et al., 2011), and body state (Brett et al., 1969; Huey & Kingsolver, 2019). We have not allowed organisms to evolve over time (Forester et al., 2022), experience acclimation (Phillips et al., 2016),

adaptation (O'Donnell et al., 2018), and heat shock (Boopathy et al., 2022), or to physiologically and behaviorally thermoregulate to decouple their body temperature from the air temperature (Kearney et al., 2009; Sears et al., 2016). We have further—and once again—extrapolated conclusions from a dataset of 38 invertebrates that has been repeatedly analyzed over the years (Deutsch et al., 2008; Duffy et al., 2022; Vasseur et al., 2014), despite (1) its biased focus on 50+ year-old lab populations of economically important pest insects spread unevenly across the globe (Frazier et al., 2006) and (2) the implicit complications of assuming that a single TPC can account for the daily changes to population growth rate of invertebrates with multiple life stages timed across seasonal cycles (Kingsolver & Buckley, 2020). While these various assumptions may each worsen or improve the predictions we have made, the general patterns of how autocorrelation mediates extinction risk will hold. Exploring whether more realistic assumptions about how organisms interact with their thermal environments exacerbate or negate these risks should be an area of critical consideration moving forward.

We have shown that given a thermal regime, extinction risk can shift dramatically due solely to changes in the temporal autocorrelation structure—even when classic metrics like average population size and TSMs indicate no risk. While the statistical quantities describing a regime's temperature distribution provide valuable information about the prevalence of stressful events when paired with an organism's TPC, the picture is not complete without considering the order in which those temperatures occur. It is thus critical to appreciate the noise structure of stressful thermal regimes when assessing extinction risks, especially as temperatures grow more stressful and heatwave regimes continue to intensify.

AUTHOR CONTRIBUTIONS

Alison J. Robey and David A. Vasseur conceptualized the study and model design, Alison J. Robey gathered data, performed analyses, and wrote the first draft, and both authors contributed to revisions.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data (Robey & Vasseur, 2025) are available in Dryad at <https://doi.org/10.5061/dryad.w0vt4b91q>. Code (Robey, 2025)

is available in Zenodo at <https://doi.org/10.5281/zenodo.17728683>. Data includes a previously published TPC dataset from Deutsch et al. (2008) (<https://doi.org/10.1073/pnas.0709472105>), reproduced from Frazier et al. (2006) (<https://doi.org/10.1086/506977>). Historical temperature time series data were retrieved from Visual Crossing Weather (Visual Crossing Corporation, 2024; <https://www.visualcrossing.com/>); query details for data retrieval are described in Appendix S4.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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